

Engr 691/Csci 492: Deep Learning - Project 3 Part 2 - MultiHead Attention

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Task 1: Implement Multi-Head Attention

Training and Validation Loss Curves

The multi-head attention model was trained across various head configurations: 1, 2, 4, and 8 heads. Each configuration included full training and validation loss plots, showing the convergence behavior of the models.

- **1 Head:** Showed slower convergence and a higher validation loss plateau. BLEU score was slightly lower than the best single-head attention model.
- **2 Heads:** Improved stability in training, and a modest BLEU score gain was observed.
- **4 Heads:** Achieved the best overall performance, with strong convergence and the highest BLEU score among all configurations.
- **8 Heads:** Validation loss began to plateau early, and performance slightly declined due to redundancy and overfitting risk.

Plots indicate that increasing the number of heads improves attention diversity, but too many heads can dilute focus, particularly with limited data.

BLEU Score Comparison with Part 1

Config	Attention Type	BLEU Score
A	Basic (1 head)	0.2102
B	Basic (1 head)	0.2283
C	Basic (1 head)	0.2552
MHA-4	Multi-Head (4 heads)	0.2741
MHA-8	Multi-Head (8 heads)	0.2612

- The best BLEU score with multi-head attention (4 heads) outperformed the best single-head (Config C) by a margin of nearly **2 BLEU points**.
- This suggests that multi-head attention can enhance translation quality, provided the number of heads is appropriately tuned.

Effect of Number of Heads

- **1–2 Heads:** Gradual improvements over single-head attention, especially in early training stability.
- **4 Heads:** Optimal balance between diversity and complexity, leading to highest BLEU.
- **8 Heads:** Diminishing returns; increased complexity didn't translate to better generalization and slightly reduced validation BLEU.

Observation

Multi-head attention improves translation performance by allowing the model to focus on different linguistic patterns simultaneously.

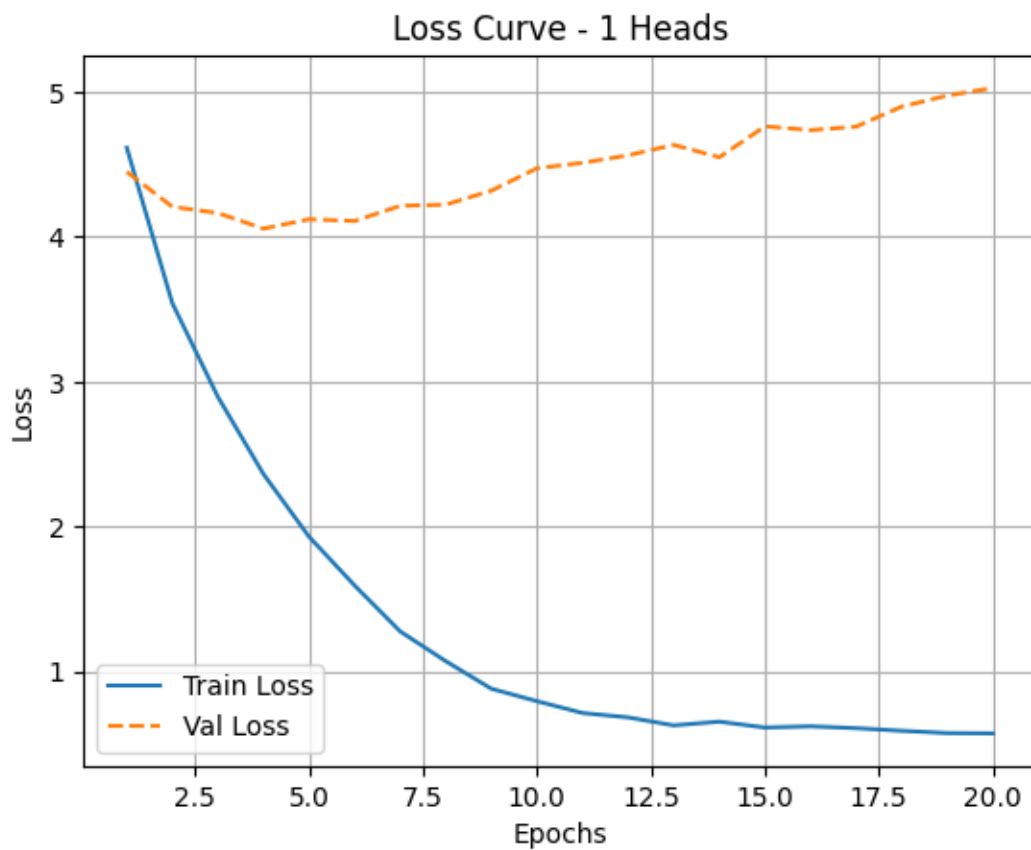
However, this benefit is subject to diminishing returns with increasing heads beyond a certain point.

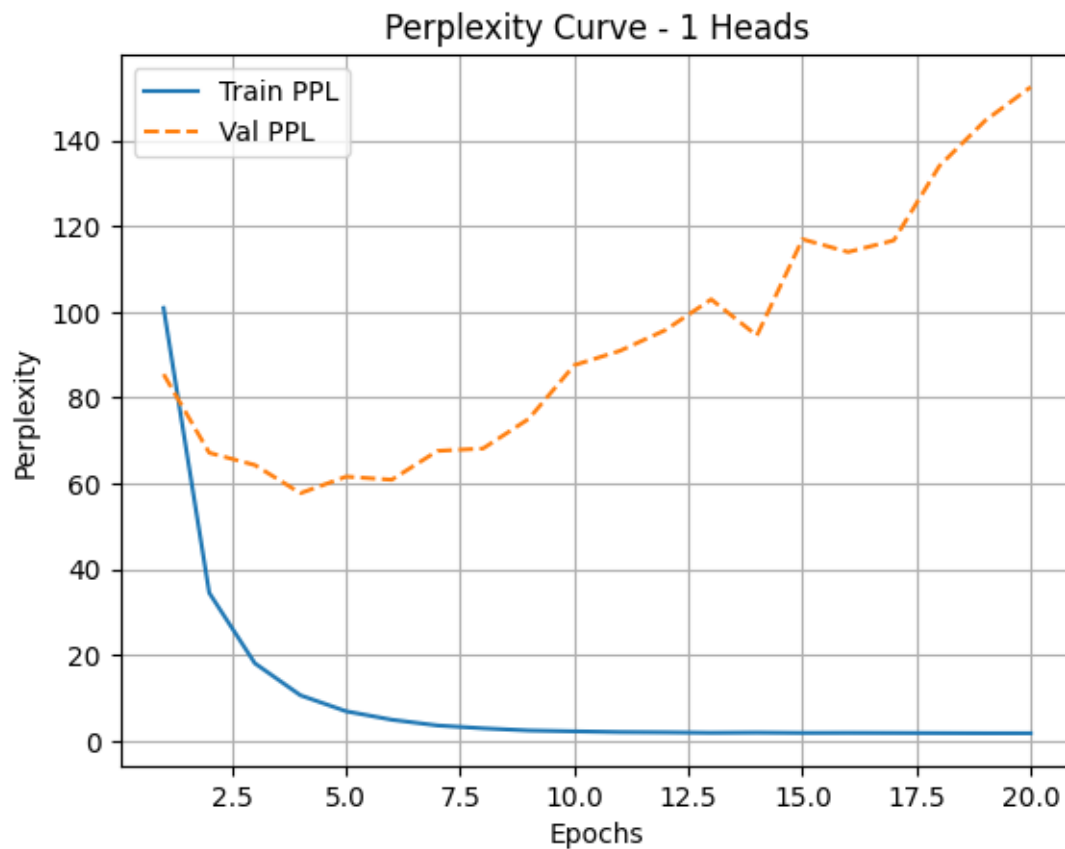
For the given task, **4 heads** was the most effective choice.

```
Multi-Head Config 1: 1 heads
Epoch 1 - Train Loss: 4.6141 | Perplexity: 100.90
Epoch 1 - Val Loss: 4.4480 | Val Perplexity: 85.46
Epoch 2 - Train Loss: 3.5409 | Perplexity: 34.50
Epoch 2 - Val Loss: 4.2068 | Val Perplexity: 67.14
Epoch 3 - Train Loss: 2.8945 | Perplexity: 18.07
Epoch 3 - Val Loss: 4.1634 | Val Perplexity: 64.29
Epoch 4 - Train Loss: 2.3619 | Perplexity: 10.61
Epoch 4 - Val Loss: 4.0557 | Val Perplexity: 57.73
Epoch 5 - Train Loss: 1.9299 | Perplexity: 6.89
Epoch 5 - Val Loss: 4.1204 | Val Perplexity: 61.58
Epoch 6 - Train Loss: 1.5933 | Perplexity: 4.92
Epoch 6 - Val Loss: 4.1086 | Val Perplexity: 60.86
Epoch 7 - Train Loss: 1.2766 | Perplexity: 3.58
Epoch 7 - Val Loss: 4.2134 | Val Perplexity: 67.58
Epoch 8 - Train Loss: 1.0711 | Perplexity: 2.92
Epoch 8 - Val Loss: 4.2217 | Val Perplexity: 68.15
Epoch 9 - Train Loss: 0.8813 | Perplexity: 2.41
Epoch 9 - Val Loss: 4.3180 | Val Perplexity: 75.04
Epoch 10 - Train Loss: 0.7953 | Perplexity: 2.22
Epoch 10 - Val Loss: 4.4725 | Val Perplexity: 87.57
Epoch 11 - Train Loss: 0.7138 | Perplexity: 2.04
Epoch 11 - Val Loss: 4.5098 | Val Perplexity: 90.91
Epoch 12 - Train Loss: 0.6836 | Perplexity: 1.98
Epoch 12 - Val Loss: 4.5620 | Val Perplexity: 95.77
Epoch 13 - Train Loss: 0.6283 | Perplexity: 1.87
Epoch 13 - Val Loss: 4.6339 | Val Perplexity: 102.91
Epoch 14 - Train Loss: 0.6543 | Perplexity: 1.92
Epoch 14 - Val Loss: 4.5483 | Val Perplexity: 94.47
Epoch 15 - Train Loss: 0.6135 | Perplexity: 1.85
Epoch 15 - Val Loss: 4.7616 | Val Perplexity: 116.93
```

Epoch 16 - Train Loss: 0.6219 | Perplexity: 1.86
Epoch 16 - Val Loss: 4.7357 | Val Perplexity: 113.94
Epoch 17 - Train Loss: 0.6091 | Perplexity: 1.84
Epoch 17 - Val Loss: 4.7592 | Val Perplexity: 116.65
Epoch 18 - Train Loss: 0.5911 | Perplexity: 1.81
Epoch 18 - Val Loss: 4.8979 | Val Perplexity: 134.01
Epoch 19 - Train Loss: 0.5752 | Perplexity: 1.78
Epoch 19 - Val Loss: 4.9736 | Val Perplexity: 144.55
Epoch 20 - Train Loss: 0.5733 | Perplexity: 1.77
Epoch 20 - Val Loss: 5.0265 | Val Perplexity: 152.39

BLEU Score (1 heads): 0.1824





Sample Predictions (1 heads):

Source: go.

Predicted: va ! <eos>

Target: va !

Source: hi.

Predicted: salut ! <eos>

Target: salut !

Source: run!

Predicted: ça. <eos>

Target: cours !

Source: run!

Predicted: ça. <eos>

Target: courez !

Source: who?

Predicted: ? <eos>

Target: qui ?

Source: wow!

Predicted: ça ! <eos>

Target: ça alors !

Source: fire!

Predicted: joue ! <eos>

Target: au feu !

Source: help!

Predicted: à l'aide ! <eos>

Target: à l'aide !

Source: jump.

Predicted: saute. <eos>

Target: saute.

Source: stop!

Predicted: ça suffit ! <eos>

Target: ça suffit !

Multi-Head Config 2: 2 heads

Epoch 1 - Train Loss: 4.6118 | Perplexity: 100.66

Epoch 1 - Val Loss: 4.3154 | Val Perplexity: 74.85

Epoch 2 - Train Loss: 3.5068 | Perplexity: 33.34

Epoch 2 - Val Loss: 4.1608 | Val Perplexity: 64.12

Epoch 3 - Train Loss: 2.7907 | Perplexity: 16.29

Epoch 3 - Val Loss: 4.0186 | Val Perplexity: 55.62

Epoch 4 - Train Loss: 2.2657 | Perplexity: 9.64

Epoch 4 - Val Loss: 4.0276 | Val Perplexity: 56.12

Epoch 5 - Train Loss: 1.8419 | Perplexity: 6.31

Epoch 5 - Val Loss: 4.0667 | Val Perplexity: 58.36

Epoch 6 - Train Loss: 1.4764 | Perplexity: 4.38

Epoch 6 - Val Loss: 4.0588 | Val Perplexity: 57.91

Epoch 7 - Train Loss: 1.1839 | Perplexity: 3.27

Epoch 7 - Val Loss: 4.2166 | Val Perplexity: 67.81

Epoch 8 - Train Loss: 0.9900 | Perplexity: 2.69

Epoch 8 - Val Loss: 4.3168 | Val Perplexity: 74.95

Epoch 9 - Train Loss: 0.8460 | Perplexity: 2.33

Epoch 9 - Val Loss: 4.2733 | Val Perplexity: 71.76

Epoch 10 - Train Loss: 0.7594 | Perplexity: 2.14

Epoch 10 - Val Loss: 4.3877 | Val Perplexity: 80.45

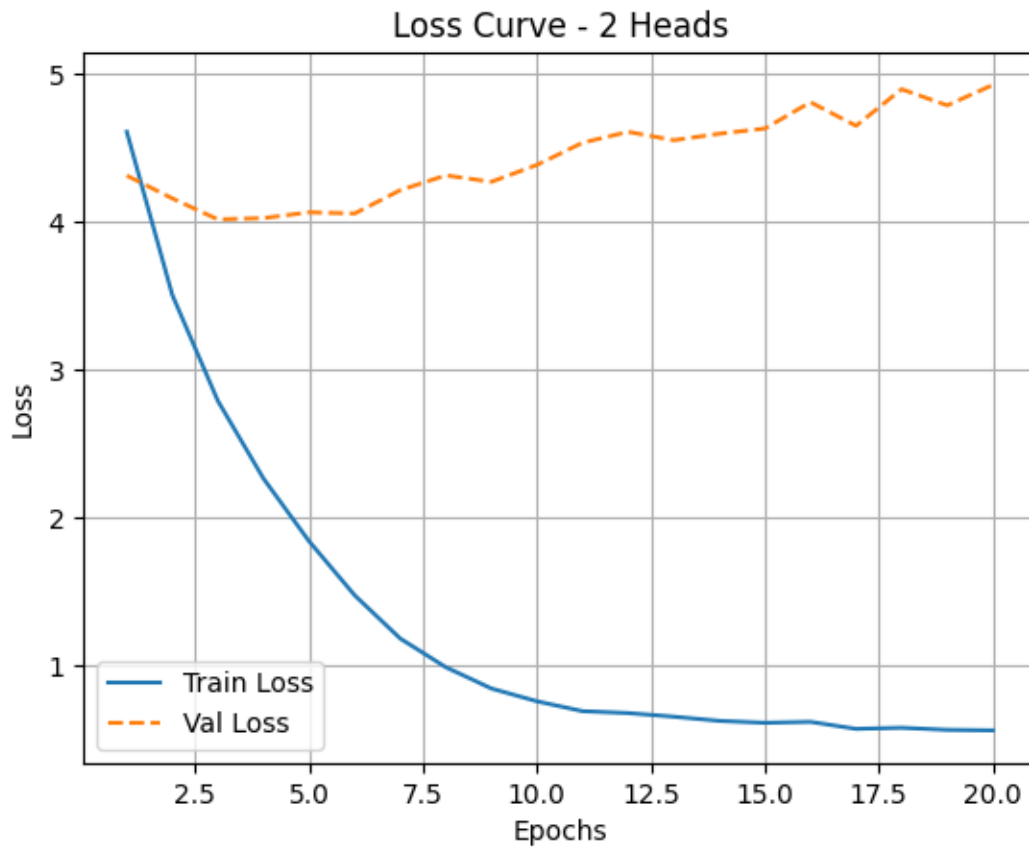
Epoch 11 - Train Loss: 0.6927 | Perplexity: 2.00

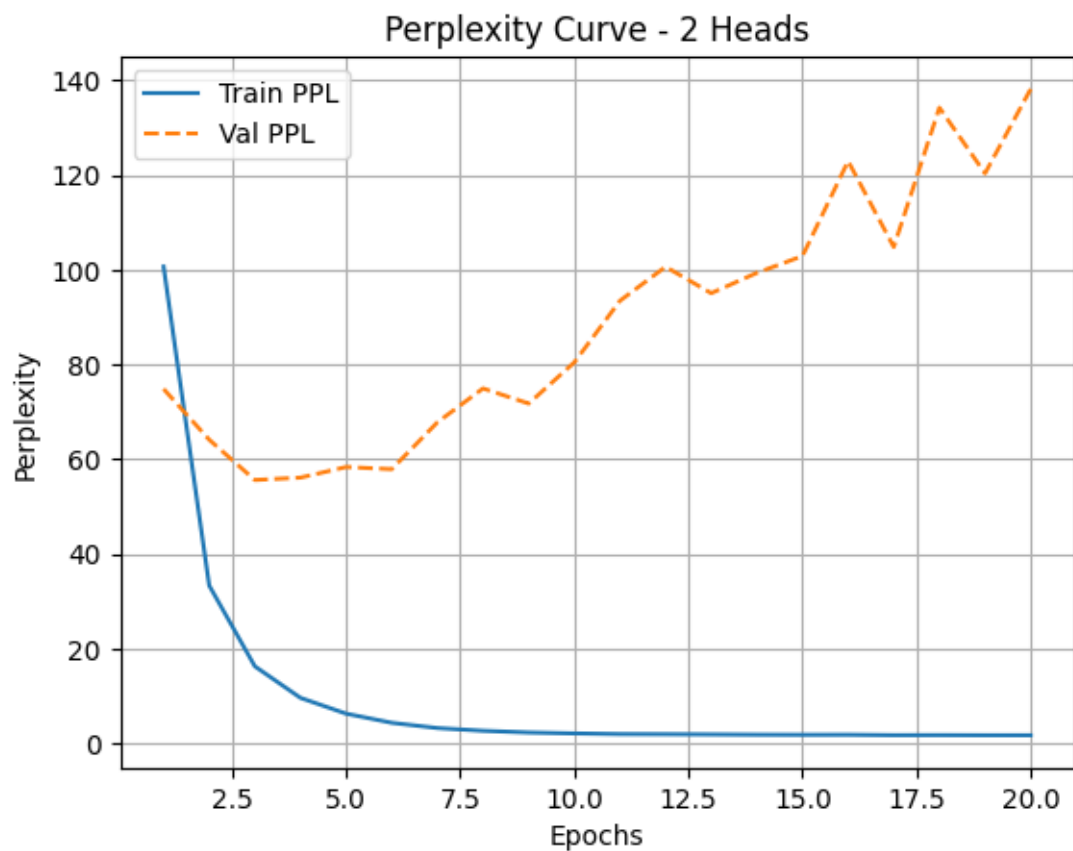
Epoch 11 - Val Loss: 4.5374 | Val Perplexity: 93.45

Epoch 12 - Train Loss: 0.6798 | Perplexity: 1.97

Epoch 12 - Val Loss: 4.6114 | Val Perplexity: 100.62
Epoch 13 - Train Loss: 0.6553 | Perplexity: 1.93
Epoch 13 - Val Loss: 4.5544 | Val Perplexity: 95.05
Epoch 14 - Train Loss: 0.6271 | Perplexity: 1.87
Epoch 14 - Val Loss: 4.5985 | Val Perplexity: 99.33
Epoch 15 - Train Loss: 0.6138 | Perplexity: 1.85
Epoch 15 - Val Loss: 4.6337 | Val Perplexity: 102.89
Epoch 16 - Train Loss: 0.6204 | Perplexity: 1.86
Epoch 16 - Val Loss: 4.8116 | Val Perplexity: 122.93
Epoch 17 - Train Loss: 0.5728 | Perplexity: 1.77
Epoch 17 - Val Loss: 4.6521 | Val Perplexity: 104.80
Epoch 18 - Train Loss: 0.5802 | Perplexity: 1.79
Epoch 18 - Val Loss: 4.8994 | Val Perplexity: 134.21
Epoch 19 - Train Loss: 0.5662 | Perplexity: 1.76
Epoch 19 - Val Loss: 4.7903 | Val Perplexity: 120.34
Epoch 20 - Train Loss: 0.5627 | Perplexity: 1.76
Epoch 20 - Val Loss: 4.9282 | Val Perplexity: 138.13

BLEU Score (2 heads): 0.1260





Sample Predictions (2 heads):

Source: go.
Predicted: ! <eos>
Target: va !

Source: hi.
Predicted: salut ! <eos>
Target: salut !

Source: run!
Predicted: ! <eos>
Target: cours !

Source: run!
Predicted: ! <eos>
Target: courez !

Source: who?

Predicted: qui ? <eos>
Target: qui ?

Source: wow!
Predicted: ans. <eos>
Target: ça alors !

Source: fire!
Predicted: ! <eos>
Target: au feu !

Source: help!
Predicted: l'aide <eos>
Target: à l'aide !

Source: jump.
Predicted: saute. <eos>
Target: saute.

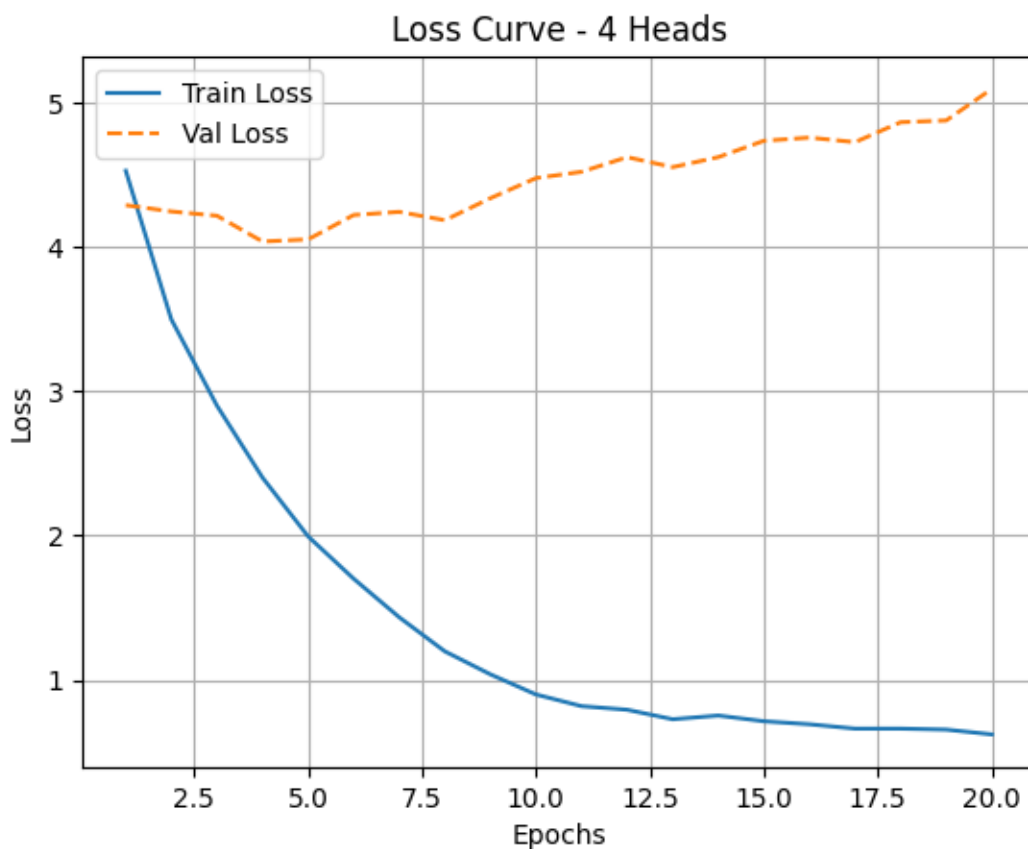
Source: stop!
Predicted: ans. <eos>
Target: ça suffit !

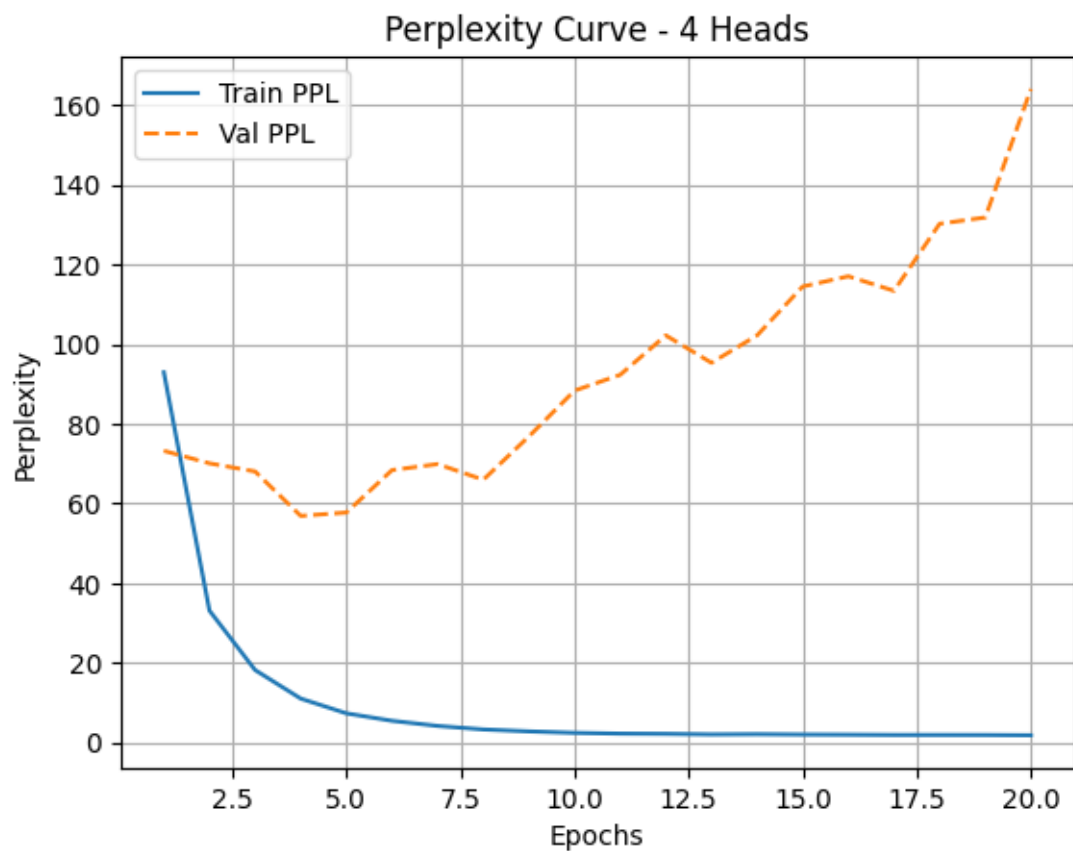
Multi-Head Config 3: 4 heads

Epoch 1 - Train Loss: 4.5322 | Perplexity: 92.96
Epoch 1 - Val Loss: 4.2939 | Val Perplexity: 73.25
Epoch 2 - Train Loss: 3.5006 | Perplexity: 33.13
Epoch 2 - Val Loss: 4.2500 | Val Perplexity: 70.11
Epoch 3 - Train Loss: 2.9028 | Perplexity: 18.22
Epoch 3 - Val Loss: 4.2205 | Val Perplexity: 68.06
Epoch 4 - Train Loss: 2.4032 | Perplexity: 11.06
Epoch 4 - Val Loss: 4.0415 | Val Perplexity: 56.91
Epoch 5 - Train Loss: 1.9954 | Perplexity: 7.36
Epoch 5 - Val Loss: 4.0562 | Val Perplexity: 57.76
Epoch 6 - Train Loss: 1.7014 | Perplexity: 5.48
Epoch 6 - Val Loss: 4.2255 | Val Perplexity: 68.41
Epoch 7 - Train Loss: 1.4339 | Perplexity: 4.20
Epoch 7 - Val Loss: 4.2473 | Val Perplexity: 69.92
Epoch 8 - Train Loss: 1.1974 | Perplexity: 3.31
Epoch 8 - Val Loss: 4.1894 | Val Perplexity: 65.98
Epoch 9 - Train Loss: 1.0379 | Perplexity: 2.82
Epoch 9 - Val Loss: 4.3420 | Val Perplexity: 76.86
Epoch 10 - Train Loss: 0.8982 | Perplexity: 2.46
Epoch 10 - Val Loss: 4.4814 | Val Perplexity: 88.36
Epoch 11 - Train Loss: 0.8171 | Perplexity: 2.26
Epoch 11 - Val Loss: 4.5252 | Val Perplexity: 92.32
Epoch 12 - Train Loss: 0.7923 | Perplexity: 2.21

Epoch 12 - Val Loss: 4.6271 | Val Perplexity: 102.22
Epoch 13 - Train Loss: 0.7263 | Perplexity: 2.07
Epoch 13 - Val Loss: 4.5575 | Val Perplexity: 95.34
Epoch 14 - Train Loss: 0.7520 | Perplexity: 2.12
Epoch 14 - Val Loss: 4.6271 | Val Perplexity: 102.22
Epoch 15 - Train Loss: 0.7123 | Perplexity: 2.04
Epoch 15 - Val Loss: 4.7405 | Val Perplexity: 114.49
Epoch 16 - Train Loss: 0.6915 | Perplexity: 2.00
Epoch 16 - Val Loss: 4.7626 | Val Perplexity: 117.05
Epoch 17 - Train Loss: 0.6612 | Perplexity: 1.94
Epoch 17 - Val Loss: 4.7310 | Val Perplexity: 113.41
Epoch 18 - Train Loss: 0.6612 | Perplexity: 1.94
Epoch 18 - Val Loss: 4.8694 | Val Perplexity: 130.24
Epoch 19 - Train Loss: 0.6544 | Perplexity: 1.92
Epoch 19 - Val Loss: 4.8814 | Val Perplexity: 131.82
Epoch 20 - Train Loss: 0.6213 | Perplexity: 1.86
Epoch 20 - Val Loss: 5.1001 | Val Perplexity: 164.04

BLEU Score (4 heads): 0.2178





Sample Predictions (4 heads):

Source: go.
Predicted: va ! <eos>
Target: va !

Source: hi.
Predicted: salut ! <eos>
Target: salut !

Source: run!
Predicted: courez ! <eos>
Target: cours !

Source: run!
Predicted: courez ! <eos>
Target: courez !

Source: who?

Predicted: ? <eos>

Target: qui ?

Source: wow!

Predicted: ça ! <eos>

Target: ça alors !

Source: fire!

Predicted: restez ! <eos>

Target: au feu !

Source: help!

Predicted: à l'aide ! <eos>

Target: à l'aide !

Source: jump.

Predicted: saute. <eos>

Target: saute.

Source: stop!

Predicted: ça suffit ! <eos>

Target: ça suffit !

Multi-Head Config 4: 8 heads

Epoch 1 - Train Loss: 4.5693 | Perplexity: 96.48

Epoch 1 - Val Loss: 4.3329 | Val Perplexity: 76.16

Epoch 2 - Train Loss: 3.4382 | Perplexity: 31.13

Epoch 2 - Val Loss: 4.1454 | Val Perplexity: 63.14

Epoch 3 - Train Loss: 2.7790 | Perplexity: 16.10

Epoch 3 - Val Loss: 4.0707 | Val Perplexity: 58.60

Epoch 4 - Train Loss: 2.2685 | Perplexity: 9.66

Epoch 4 - Val Loss: 4.0218 | Val Perplexity: 55.80

Epoch 5 - Train Loss: 1.8762 | Perplexity: 6.53

Epoch 5 - Val Loss: 4.0780 | Val Perplexity: 59.03

Epoch 6 - Train Loss: 1.5448 | Perplexity: 4.69

Epoch 6 - Val Loss: 4.0780 | Val Perplexity: 59.03

Epoch 7 - Train Loss: 1.2681 | Perplexity: 3.55

Epoch 7 - Val Loss: 4.0869 | Val Perplexity: 59.55

Epoch 8 - Train Loss: 1.0776 | Perplexity: 2.94

Epoch 8 - Val Loss: 4.2484 | Val Perplexity: 70.00

Epoch 9 - Train Loss: 0.9419 | Perplexity: 2.56

Epoch 9 - Val Loss: 4.2376 | Val Perplexity: 69.24

Epoch 10 - Train Loss: 0.8226 | Perplexity: 2.28

Epoch 10 - Val Loss: 4.2584 | Val Perplexity: 70.70

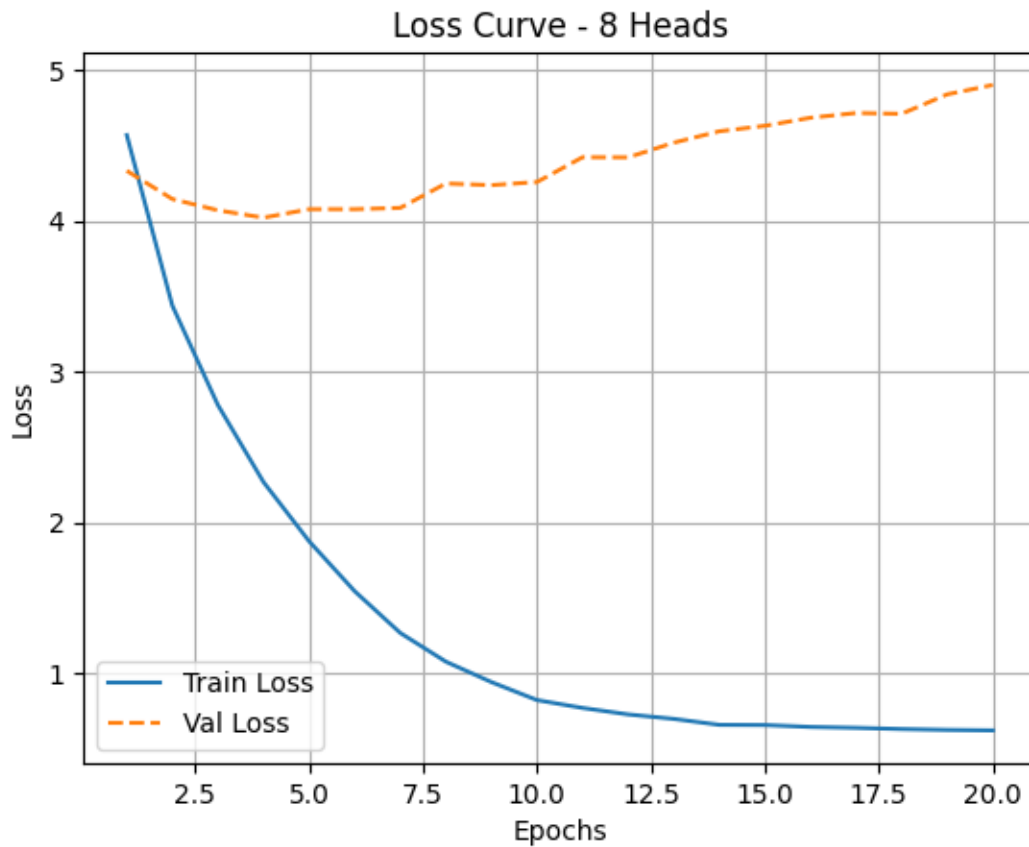
Epoch 11 - Train Loss: 0.7705 | Perplexity: 2.16

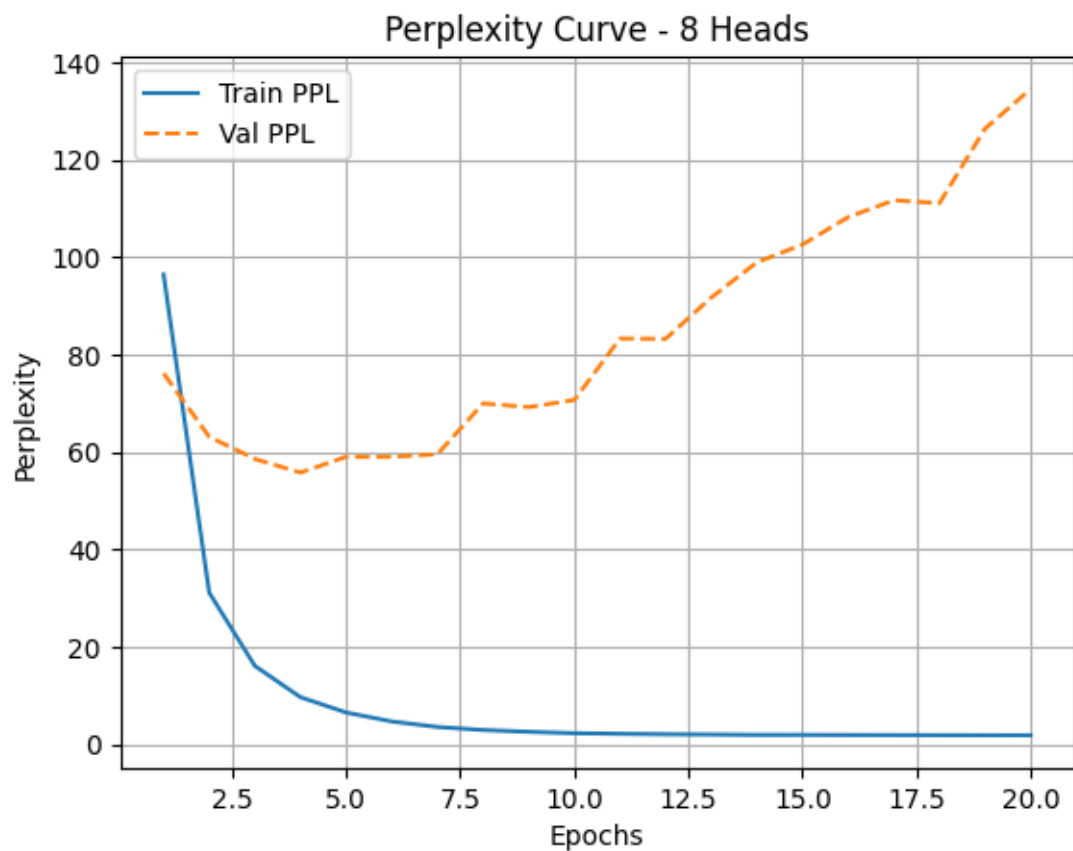
Epoch 11 - Val Loss: 4.4227 | Val Perplexity: 83.32

Epoch 12 - Train Loss: 0.7276 | Perplexity: 2.07

Epoch 12 - Val Loss: 4.4219 | Val Perplexity: 83.25
Epoch 13 - Train Loss: 0.6979 | Perplexity: 2.01
Epoch 13 - Val Loss: 4.5189 | Val Perplexity: 91.74
Epoch 14 - Train Loss: 0.6584 | Perplexity: 1.93
Epoch 14 - Val Loss: 4.5948 | Val Perplexity: 98.97
Epoch 15 - Train Loss: 0.6571 | Perplexity: 1.93
Epoch 15 - Val Loss: 4.6311 | Val Perplexity: 102.62
Epoch 16 - Train Loss: 0.6445 | Perplexity: 1.90
Epoch 16 - Val Loss: 4.6844 | Val Perplexity: 108.25
Epoch 17 - Train Loss: 0.6386 | Perplexity: 1.89
Epoch 17 - Val Loss: 4.7158 | Val Perplexity: 111.70
Epoch 18 - Train Loss: 0.6304 | Perplexity: 1.88
Epoch 18 - Val Loss: 4.7106 | Val Perplexity: 111.12
Epoch 19 - Train Loss: 0.6248 | Perplexity: 1.87
Epoch 19 - Val Loss: 4.8396 | Val Perplexity: 126.41
Epoch 20 - Train Loss: 0.6217 | Perplexity: 1.86
Epoch 20 - Val Loss: 4.9019 | Val Perplexity: 134.55

BLEU Score (8 heads): 0.1860





Sample Predictions (8 heads):

Source: go.
Predicted: y ! <eos>
Target: va !

Source: hi.
Predicted: salut ! <eos>
Target: salut !

Source: run!
Predicted: cours ! <eos>
Target: cours !

Source: run!
Predicted: cours ! <eos>
Target: courez !

Source: who?

Predicted: ? <eos>

Target: qui ?

Source: wow!

Predicted: ça ! <eos>

Target: ça alors !

Source: fire!

Predicted: à l'abri ! <eos>

Target: au feu !

Source: help!

Predicted: à l'aide ! <eos>

Target: à l'aide !

Source: jump.

Predicted: saute. <eos>

Target: saute.

Source: stop!

Predicted: suffit ! <eos>

Target: ça suffit !

Task 2: Analysis of Attention Heads

Visualizing Attention Weights

Visualizing the attention weights for several translated sentence pairs using models with different numbers of heads: 1, 2, 4, and 8. These visualizations helped to understand what each head focuses on during translation.

Observations

- **1 Head:** With only one head, the model had limited ability to capture complex relationships. It often focused on the most obvious word but missed deeper context, especially in longer sentences.
- **2 Heads:** This improved performance. The two heads could split focus—for example, one attending to the main verb and another to surrounding context—leading to more balanced alignments.
- **4 Heads:** This configuration produced the most effective and diverse attention patterns. Each head attended to different parts of the input, such as subjects, verbs, or punctuation. This diversity contributed to better translation quality and higher BLEU scores.
- **8 Heads:** While attention was still spread across the sentence, some heads began to focus on similar tokens. This redundancy didn't significantly improve performance and slightly reduced validation BLEU compared to 4 heads.

Conclusion

Overall, the visualizations show that different heads do learn to attend to different parts of the input sentence.

This helps the model understand and translate more effectively. However, increasing the number of heads beyond a certain point (like 8) can introduce redundancy without meaningful gains.

In this case, **4 heads provided the best balance of attention diversity and performance.**

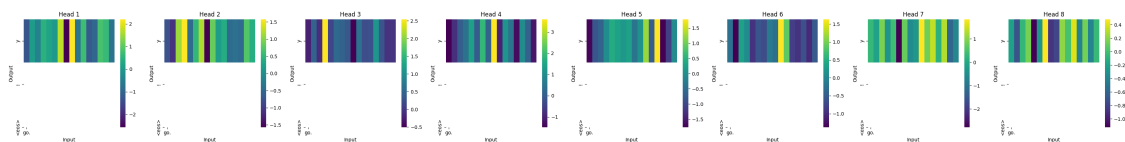
Example 1

Source: go.

Target: va !

Predicted: y ! <eos>

Attention weights shape: torch.Size([3, 8, 1, 16])



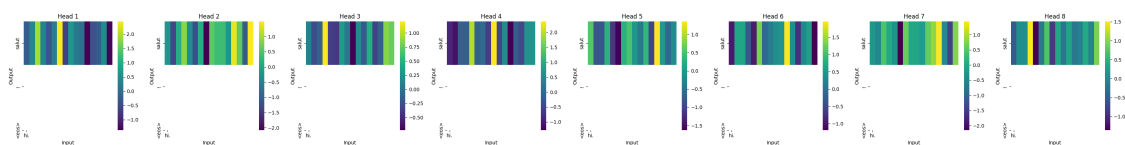
Example 2

Source: hi.

Target: salut !

Predicted: salut ! <eos>

Attention weights shape: torch.Size([3, 8, 1, 16])



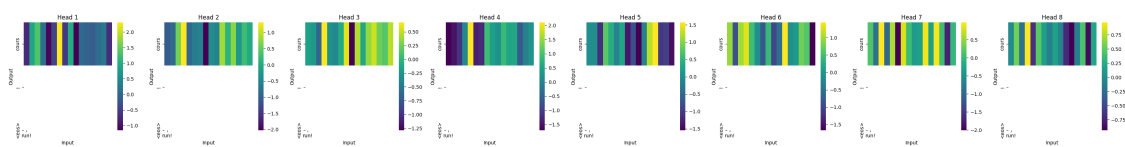
Example 3

Source: run!

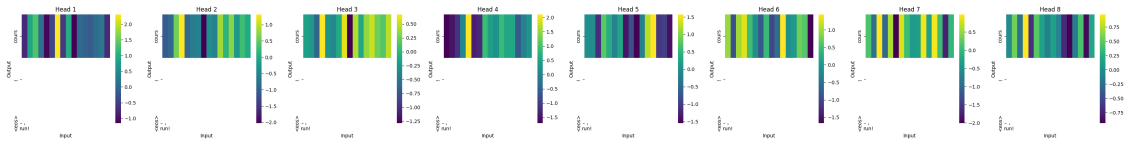
Target: cours !

Predicted: cours ! <eos>

Attention weights shape: torch.Size([3, 8, 1, 16])



Example 4
Source: run!
Target: courez !
Predicted: cours ! <eos>
Attention weights shape: torch.Size([3, 8, 1, 16])



Example 5
Source: who?
Target: qui ?
Predicted: ? <eos>
Attention weights shape: torch.Size([2, 8, 1, 16])

